Ellinor Romin, Iris Landar Lygren, Maria Eilersen

Costume design sketches for *Livet på Landet* (1943)

Introduction and background

Digitisation has become an important part of how collections are managed in the cultural heritage sector. Documents and objects that previously only were accessed physically at cultural heritage institutions are now transferred into digital formats and representations (Björk, 2015, p.1). According to Conway (2015, p. 52) most archives are either digitizing their collections or wish to do so. Further Conaway argues (2015, p. 52) that if analog sources were not available in digital form, most users would not have access to the information.

Within the field of film, digital technology has made it possible to restore and preserve film in digital formats (Lucia & Hamid, 2017, pp 339- 340). It can be seen as an important element in the preservation of film, which in itself often consists of very fragile material. A lot of film history has disappeared before it could ever be preserved (Moriarty, 2024, p. 7). In a Swedish context within the cultural heritage sector, the Swedish Film Institute is reproducing Swedish films digitally for preservation. In 2023 the Swedish Film Institute changed terminology from digitalization to digital reproduction, whereas digital reproduction refers to the complex interpretation process from analog to a digital form, digitisation refers to machine transferring such as scanning (Svenska Filminstitutet n.d).

'Svenska Filminstitutets bildarkiv' is an archive within the Swedish Film Institute. The archive has a collection of stills and posters from most of the films that have been screened in Swedish cinemas, from the early days of film in the 20th century to present times. The archive also has a collection of portraits of film personalities, older Stockholm cinemas, various film documents and technical documentation (Svenska Filminstitutet n.d).

The material

The material we chose for our project, costume design sketches, is located at 'Svenska Filminstitutets bildarkiv' (The Swedish Film Institute image archive). The sketches were created for the production of the film *Livet på landet,* a Swedish feature film from 1943, directed by Bror Bügler and produced by *AB Europa Film*. The film is an adaptation of Fritz Reuter's *Ut mine Stromtid*, a trilogy of novels that were published in Swedish translation 1870. Several film adaptations have been made of the novel trilogy, the first Swedish film adaption was made in 1924. (Svensk Filmdatabas n.d).

Edvard Persson played the main character in the film, *Onkel Bräsig*. Edvard Persson was very popular and a public phenomenon. Large parts of the Swedish cinema audience watched Swedish films during the war years and most watched films featuring Edvard Persson. *Livet på landet* was a huge success and was seen by an audience of a million (Furhammar, 1991 p.185-186).

The costume design sketches are made by the film's architect Max Linder. The sketches are made on relative thick art paper with aquarelle and/or pencil. The sketches have been preserved quite well, although the paper has turned yellow and on some of the sketches there are moisture spots, small holes and some color smudge.

Method

Selection

We contacted 'Svenska Filminstitutets bildarkiv' to present our project. We described the premises regarding copyright and our intention to publish the material in an educational context. At our first visit, an archivist showed us some material he had in mind that was cleared to use regarding copyright. It was also decided that we would use the inhouse scanners belonging to the archive for the image capture.

The first material we were shown was a catalogue of extras from the 1950s with stills and information about the background actors. The second material was stills from old cinemas in Stockholm. One selection that, unfortunately, we could not use due to the format of the sketches being too large for the available scanners, was the costume design sketches to the classic film "Fanny and Alexander" by Ingmar Bergman.

However, we found other costume sketches that fit both the purpose of our digitization project as well as the inhouse scanner. This small selection consisted of 40 costume sketches from seven different films, released between 1940-1945, produced by *AB Europafilm*. These costume design sketches had never been digitized before.

The selection of material was made out of two criteria: It should be a unique material of some kind that hadn't been digitized before, and the copyright should also be cleared for our purposes. Based on Cowick's priority list (2018, p.12), the score we get retrospectively is a medium priority item. The sketches have some historical/research value, no high demand, somewhat fragile, no legal issues and not otherwise available online.

We made an agreement with the archive to digitize these 40 sketches. To make the amount of material more manageable within the framework of our project, we later made the decision to limit the digitization project to sketches from one of these seven films: *Livet på Landet.* The material included 19 costume design sketches, a text document and a business card with handwriting.

Image capture

The image capture was done over the course of two days at the Swedish Film Institute library*,* on March 26th and 27th 2026. We used the archive's two scanners, located in the library's research room.

According to the categories Cowick (2018, p. 59) lists for different items for digitization, the sketches can be defined as Art. For this reason, the preparations we made were very similar to the guidelines of a digitization workflow for in-house digitization, suggested by Cowick (2018, p. 61). As our project is based on images rather than text, we decided to have a high quality of the images. The resolution was set to 600 ppi with 48-bit color. The master files were saved in a TIFF format, and the original size of the sheets kept (”måttstorlek: original”).

The main scanner used was the Epson Perfection V750 pro, but for particularly large images we used the bigger scanner, the Expression 12000xl. For the files from *Livet på Landet*, the images that we ended up using on the web page were primarily scanned with the former. Only the instructive text sheet, *livet_pa_landet_20,* was scanned using the latter, due to its size.

When handling each sketch, protective gloves were used. We inspected each sheet thoroughly, and used an air duster to remove any dirt. We also wiped the glass of the scanner with a microfiber cloth, as well as using the air duster.

After capturing the first few sketches, we examined the lighting conditions in the room and compared the image of the scanned sketch with the original. The adjustments we made was to turn off an automation of the contrast on the scanner. We also had to manually turn off the color correction feature between each sketch.

After having scanned a large selection of the material, we discovered that the scanner had cropped part of the images. The DFG Guidelines say to store photographs in their entirety, “with a thin surrounding edge to indicate that nothing has been cut from the original” (2013, p. 22). Because of this, we changed a pre-set-up on the scanner and once we’d learned how to place the sheets on the scanner without the image being cropped, we started over, scanning each sketch once again, this time thoroughly inspecting that there was a bit of margin around the edges of the sheet.

Once each sheet had been captured, we went through them again to make sure everything looked good: we made sure there was no blurry part on the images, that there were margins around the sheet, and that the colour grading looked more or less similar between the images.

During a supervision session with our project supervisor we found that the material aspects of the sketches were of great importance. We realized we had omitted to measure the original sketches. Therefore, one of us revisited the archive to make additions; to measure the size of the original sketches and examine the conditions and the materials more carefully. These findings were then added to the metadata and descriptions.

Metadata

In our search for a suited controlled vocabulary we chose the *Getty Art & Architecture Thesaurus* for describing items of art and material culture. For describing film-related material, we chose the vocabulary *Svenska ämnesord*, which is the vocabulary used by the Swedish film institute library. The fact that we found it necessary to combine vocabularies reflect a wider problem within existing library and organization systems in relation to film. Traditional classification systems have historically been difficult to use in regards to film as a medium (Higgins, 2022, pp. 80-81). This could be partly since the systems were originally designed for books and not audiovisual media. That the Swedish Film Institute library uses Svenska ämnesord in combination with their own classification system, rather than for example DDC, could be seen as a reflection of this.

While using only one vocabulary, or two vocabularies in the same language, would have been preferable, there was no controlled vocabulary that could cover the aspects we’d like to include. For this reason, a combination of these two was best suited for covering the metadata in our project.

Dappert et al. (2016 p. 3) defines the concept *metadata profile* as a set of standards that are customized to the needs of the project. Our decision to use two standards of controlled vocabularies is thus a deliberate strategy and can be seen as part of our metadata profile. Since our project is mainly based on images we decided not to mark up the text. Therefore, we instead included a relatively detailed description, including transcription, in the metadata.

Image processing

As stated regarding the image capture we chose to use TIFF format for the masterfiles to maintain high quality. To display the images on the web the files were too large. We had to make edited copies of the masterfiles with as high resolution as possible to keep details such as texture and condition, but still small enough in file size to load without complications on GitHub. To display them in a fair way we also cropped and aligned them where needed. For this we used *Adobe Photoshop 2026* where the images were first aligned and secondly cropped. As most of the images were the same format in physical form we cropped all of those images to the same exact size. The two images with different formats were cropped to sizes matching the others as much as possible. The images were cropped using W x H x Resolution to ensure kept quality. Since we had chosen to display the images in JPG-format on the GitHub-page, they were converted to JPG after the metadata and cropping were already set. This order was chosen to make sure all the data would transfer consistently. The JPG-files were converted with as high quality as possible, 7, and the image profile set to sRGB to ensure the colors will be displayed as equivalent as possible on different screens and through different web browsers. These files have large enough quality to display the documents on the web, but still were too large to load satisfactory as thumbnails. Therefore we made versions of the JPG-images that were cropped to an even smaller size, but still with high quality conversion and metadata intact, to use as thumbnails.

Preservation

For preservation and long-term usability Dappert et al. (2016, p. 2) means that it’s necessary to gather enough information so it can be accessible in the future. Which information is needed for the metadata to be used and understood in a long term? Our decision to use established controlled vocabulary within the fields of both art and film, albeit separately, will hopefully make the metadata understood and usable in the long term.

Cowick (2018, p. 36) lists criteria recommended for preservation regarding file formats and metadata. We've used TIFF, which allows inclusion of metadata fields within the files. Another recommendation for preservation is multiple copies. The master files for our project are stored on a hard drive, a USB flash drive and on a cloud storage. The link to the cloud storage on Google Drive is available in the GitHub repository on the website. The Swedish Film Institute also received the files we created during the image capturing.

Web Page

The web page was created using HTML/CSS and Javascript, in accordance to instructions from the course as well as the DFG Guidelines, which list both HTML and CSS as suitable formatting languages (2013, p. 34). Along with a gallery page, where all the sketches can be viewed in a smaller format, each sketch has its own page with a zoom function (inspired by a previous project from this program, *Santessons Skor)* and metadata containing image descriptions as well as basic information such as copyright and artist.

The zoom function on each sketch's page was made using javascript. On hover, the mouse zooms on the picture, similar to using a magnifying glass. This, in combination with the image's high resolution, makes visible several details that might otherwise have gone unnoticed, such as light pencil strokes and small stains of colour.

We have prioritized making the zoom function available for the costume sketches, rather than the few sheets of text. In further building of the website, we would make the zoom function more extensive – see *Results and Analysis* below.

Furthermore, availability of the page was a large focus, which made us strive to code the HTML accordingly. Every img-file has an alt description referring to the metadata description, and we strived to use the correct HTML5 sectioning (<main>, <nav>, <section>, etc.) rather than merely <div>. The image gallery, navigation bar, landing page, and about page, are all responsive using CSS Flexbox. In order to keep the zoom function properly working, the image size had to be hard coded using specific pixel size, which made us unable to use CSS Flex. However, the layout of the page still allows for *some* variation of screen size, albeit not a small screen.

The web page was deployed using GitHub Pages. Once the page was deployed, the various paths to the images and HTML pages had to be altered to avoid a 404 error, and the JavaScript zoom function began to malfunction, while working on reload. Due to timing constraints, as of writing this report this has not been resolved.

Results and Analysis

In our project, we've aimed to present representations of the costume sketches in a way that resembles the originals as much as possible. We wanted to capture the material aspects of the sketches. From a user perspective, we've deliberately chosen a high resolution and a zoom-function to make it possible to take a close look at details. Even so there are still aspects that cannot be fully captured in the transfer from analogue to digital.

Milekics (2017) means that the tactile aspects are easily lost in interfaces with digital collections. On the one hand, how we would have done differently in terms of the interface is difficult to assess. On the other hand, we would have liked to include more parts of the film history context on our web page. Our initial plan was to link the costume design sketches to images of the actual costumes made for the film. Along with stills from the film with the characters wearing the costumes, it is likely that this would have added a sense of how the sketches were part of a film production and an artistic idea from sketch to finished result. Unfortunately it was not possible for us to include this extra material within the project's time frame.

The biggest takeaway from the digital process has been that we ought to have deployed the page (using GitHub Pages or other available deployment software) at an earlier stage in the process. This might have allowed for us to solve any ongoing issues, such as the malfunctioning zoom function. It would also have allowed for us to send the work in progress to the Swedish Film Institute to receive feedback. The reason we did not do this, was that the Swedish Film Institute made it clear during our visit in March that while they were grateful to have the sketches scanned and digitized, they would not be using our end product (a k a, the website). As stated above, they received the scanned images on their own computers, and have been available for questions about the movies and copyrights, but the collaboration has not gone further than that.

Another function we would have liked to have extended further is the page's zoom function. It was made to work on one image per HTML page, which created an issue on those pages displaying two images. Those pages are sketch07.html ("Marie möllers bäste dräkt") and sketch19.html ("stövel till dräkt"), where the decision was made to display sketches next to their cover page or back side, respectively, with the zoom function being displayed below the metadata. With unlimited time, we would have extended the zoom function to work on multiple images per page, as well as added a toggle button to turn it on and off. (Such a button might also have facilitated the function working on the deployed site, considering that the zoom often works on reload.)

Other than that, there are numerous functions that could have been added. A slideshow that allows the user to slide between images would have made the page more interactive and more similar to handling physical material, considering Milekic's opinion about the tactile aspects being lost in handling a digital copy. We also could have added further navigation options on the web page, in accordance with the DFG Guidelines (2013, p. 41), such as adding a search function or back- and forward buttons, for easier navigation between each sketch.

It's obvious that even in our small-scale digitizing project, there were several different decisions in each step of the process. The result of the digital representations we've digitized could have been different if we'd made different decisions, including decisions at a detailed level. For example, if we'd decided to make a slight adjustment of the colors in the image processing, the representation would have looked different. Most likely, a user would think that the original would have had the same color as the representation with the adjusted color.

An interesting aspect in regards to film and the colors represented in the sketches that could be further explored, if we had more time at hand, is how these colors were captured by the camera. In added notes we can see that clothing which might be interpreted as black have been specifically asked to not be produced in black. This might be because black is a color usually avoided in filmmaking since texture might disappear in the resulting image. The film is also produced in black and white which raises questions about how the production have reasoned regarding the clothes color choices. Color does not reflect in the same way on black and white film as it does in color film. For example: blue can translate as white and red often translates as nearly black in old black and white film due to how the light is absorbed. For this reason green was often used instead of red in makeup, and makeup became important since the actors' faces might not register well on film without it in black and white (Bordwell, Thompson & Smith, 2020, pp. 121-122). Still, the way actors feel might reflect in the acting they do and therefore some coloring might be kept on set for that reason as well. How the film's creators reflected on the coloring in this case is not evident in the materials we analysed, but could be an interesting aspect to analyze further if we had the time. For instance, how does the green costume of the character “the hunter” present in the actual film?

Dahlström & Hansson (2019, p. 4) means the digitizing process might create a distance between the original document and the digitized representation. If there’s a distance, the authenticity can be questioned and that affects the reliability of the object. Dahlström & Hansson (2019, p. 5) suggests that the user is provided a link to the uncompressed raw files.

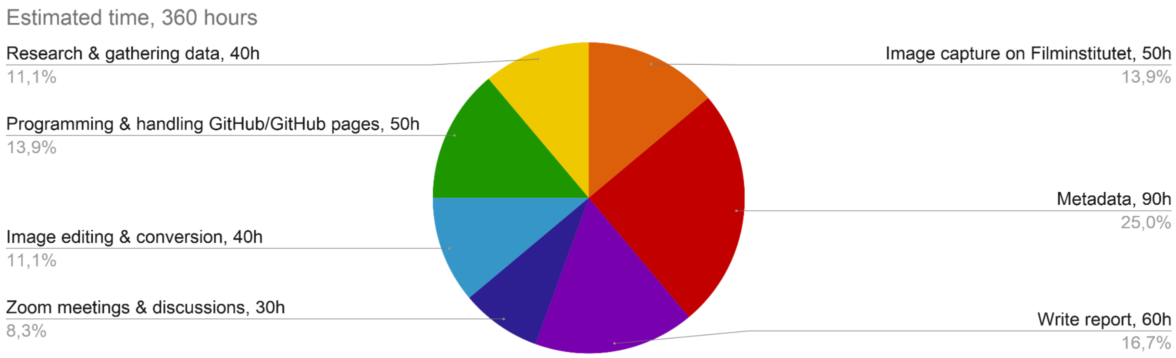
The ability for a user to see an uncompressed masterfile is of course a way to bridge the gap to the original, not only for a researcher, but for all kinds of users, it provides a kind of transparency of the digitization process. Despite this, as we've now experienced, there are several steps earlier in the process, not only in the image processing where choices might affect the digitized representation. The question is whether it’s possible to completely overcome the uncertainty regarding authenticity in a digitized representation?
Antoniazzi (2021) opens his article on digital film heritage by stating that media heritage is fundamental for understanding modern society, both historically and aesthetically. Film preservation affects the ability to experience cultural memories. However, Antoniazzi goes on, this is a mindset standing in conflict with IT companies creating systems with deliberately short lifespans. This makes it near impossible to plan for the long-term preservation of artefacts, both from the film heritage sector and other areas.

As stated above, our project uses HTML/CSS and vanilla JavaScript – that is, JavaScript that is not used within an external framework like React or Vite – because this is the fundamental basis of the internet. We have, however, used GitHub, which is an external function that could, theoretically, stop working one day, or become locked behind a paywall. As of today, the odds of this happening could be considered slim, but in a world where capitalism runs the IT factor, it is not an impossible thought.

There is also something that could be said about this being a school project. Any attempts to keep this website on the web, will be dependent on the individual people in this group and their ability to keep the website on the web, five, ten, twenty years from now.

Even so, the Swedish Film Institute has received the digital copies made, and the Google Drive is available on the project's GitHub page for anyone to download. Should the website disappear, the individual facsimiles will still exist, both with and without metadata.

We estimated the time the project took to be around 360 hours. That is around 120 hours each. This was estimated from the numbers in the following diagram.



References

Antoniazzi, L. (2021). Digital preservation and the sustainability of film heritage. Information, Communication & Society, 24(11), 1658–1673. https://doi-org.lib.costello.pub.hb.se/10.1080/1369118X.2020.1716042

Björk, L. (2015) *How reproductive is a reproduction. Digital transmission of text-based documents*. University of Borås

Bordwell, D., Thompson, K., & Smith, J. C. (2020). *Film art : an introduction* (Twelfth edition, International student edition). McGraw-Hill Education.

Cowick, C. (2018). *Digital curation projects made easy : A step-by-step guide for libraries, archives, and museums*. Bloomsbury Publishing USA.

Conway, P. (2015). Digital transformations and the archival nature of surrogates. *Archival Science, 15*(1), 51-69. https://doi.org/10.1007/s10502-014-9219-z

Dalstrom, M., & Hansson, J. (2019). Documentary Provenance and Digitized Collections: Concepts and Problems. *Proceedings from the Document Academy*, *6*(1). https://doi.org/10.35492/docam/6/1/8

Dappert, Angela., Guenther, Rebecca., & Peyrard, Sébastien. (Eds.). (2016). *Digital Preservation Metadata for Practitioners : Implementing PREMIS* (1st ed. 2016.). Springer International Publishing. https://doi.org/10.1007/978-3-319-43763-7

Deutsche Forschungsgemeinschaft (2013). *Practical Guidelines on Digitisation*. https://www.dfg.de/resource/blob/176110/76abec10bdc30b41f18695145003d6db/12-151-v1216-en-data.pdf

Furhammar, L. & Svenska filminstitutet. (1991). *Filmen i Sverige: en historia i tio kapitel*. Bra Böcker i samarbete med Filminstitutet.

Higgins, C. (2022). 'I Coulda Had Class' – The Difficulties of Classifying Film in Library of Congress Classification and Dewey Decimal Classification. *Knowledge Organization*, 49(2), 79. https://doi.org/10.5771/0943-7444-2022-2-79

Lucia, C., & Hamid, R. (Eds.). (2017). *Cineaste on film criticism, programming, and preservation in the new millennium*. University of Texas Press.

Moriarty, P. (2024). The Founding of Film in Museums: The Influence of Museum Philosophy on the Creation of Their First Film Libraries and Archives. *International Journal of Librarianship*, 9(2), 4–16. https://doi.org/10.23974/ijol.2024.vol9.2.358